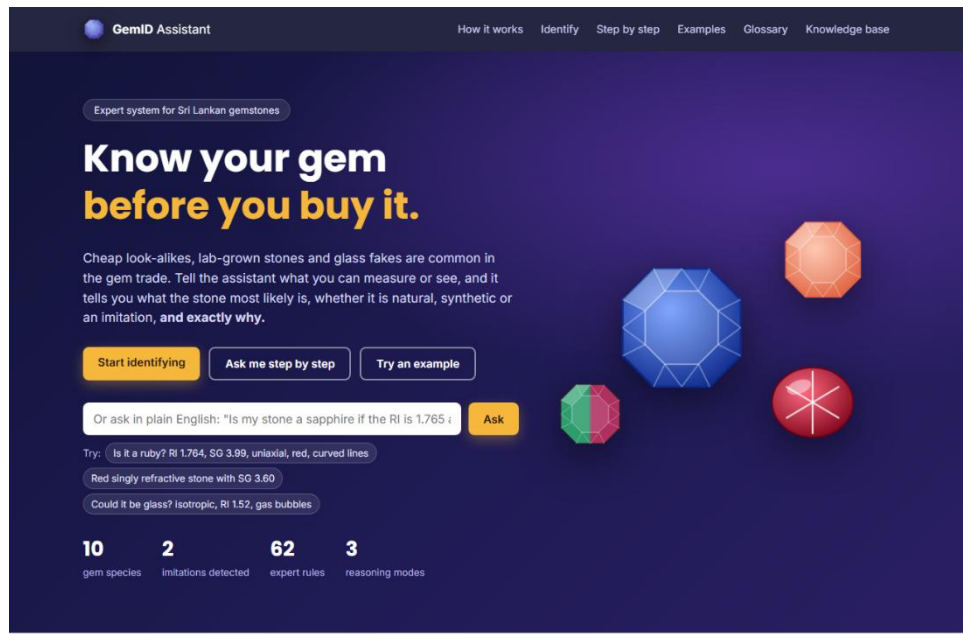


Gemstone Identification Assistant

User Manual

Installation, operation and worked examples



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1. About the system

The Gemstone Identification Assistant is an expert system that helps buyers, students and small traders identify common Sri Lankan gemstones. You describe a stone (its measurements and what you can see) and the system tells you:

- what the stone most likely is (species and variety, for example *corundum - blue sapphire*);
- whether it is likely natural, synthetic (laboratory-grown) or an imitation such as glass;
- how certain it is, as a percentage;
- what you should do next, for example obtain a laboratory certificate before buying;
- exactly how it reached its answer.

It knows ten gem species (corundum, spinel, garnet, chrysoberyl, zircon, topaz, tourmaline, quartz, feldspar/moonstone and beryl), their main varieties, and two imitation materials (glass and cubic zirconia).

Important: the system gives a first opinion only. It does not replace a certificate from the National Gem and Jewellery Authority (NGJA) or another accredited gem laboratory. Always obtain a certificate before a significant purchase.

1.1 Three ways to use it

Mode	Use it when...	What it does
Identify	"I don't know what this stone is."	Takes everything you enter and works forwards to the most likely answer (forward chaining).
Verify	"The seller says it's a ruby. Is it?"	Starts from the claim and checks every condition it needs; if the claim fails it tells you exactly why (backward chaining).
Step by step	"I want to be guided."	Asks you one question at a time, only when needed, and explains why it is asking.

You can also type a question in plain English, for example *"Is my stone a sapphire if the RI is 1.765 and it is blue?"*

2. What you need

Table 1. Requirements

Item	Requirement
Computer	Windows 10 or 11, macOS or Linux, with about 100 MB of free disk space.
SWI-Prolog	Version 9 or later (free). The system was developed and tested with SWI-Prolog 10.1.16 (64-bit).
Web browser	Any modern browser: Microsoft Edge, Google Chrome, Firefox or Safari.
Project files	The <code>gem_expert</code> folder.
Internet	Only for downloading SWI-Prolog. The system itself runs offline; without internet the page simply uses standard fonts.
Gem instruments (optional)	Refractometer, hydrostatic balance, polariscope and a 10x loupe give the best results, but the system also works with only what you can see.

3. Installation

3.1 Install SWI-Prolog

1. Open <https://www.swi-prolog.org/download/stable> in your browser.
2. Download the installer for your computer. On Windows choose the **64-bit** installer (`swipl-...-1.x64.exe`).
3. Run the installer and accept the default settings.
4. **Windows only:** on the "Install Options" page select "**Add swipl to the system PATH for all users**" (or "for the current user"). This lets you start SWI-Prolog from a terminal.
5. Finish the installation. On macOS you can alternatively install with Homebrew (`brew install swi-prolog`); on Ubuntu Linux use `sudo apt install swi-prolog`.

3.2 Copy the project folder

1. Copy or unzip the `gem_expert` folder to a convenient place, for example `Documents\gem_expert`.
2. Check that the folder contains these files: `main.pl`, `knowledge_base.pl`, `engine.pl`, `ui.pl`, `console.pl`, `diagrams.pl`, `examples.pl`, `query_parser.pl`, `tests.pl` and an `images` folder.

3.3 Check the installation

1. Open a terminal: on Windows, open the `gem_expert` folder in File Explorer, click the address bar, type `cmd` and press Enter. On macOS or Linux, open Terminal and use `cd` to go to the folder.
2. Type `swipl --version` and press Enter. You should see a line such as `SWI-Prolog version 10.1.16 for x64-win64`.

If you see "'swipl' is not recognized", SWI-Prolog is not on the PATH. See Troubleshooting (Section 15).

4. Starting and stopping the system

4.1 Start the web interface (recommended)

1. In the terminal, inside the `gem_expert` folder, type:

```
swipl main.pl
```

You will see:

```
=== Gemstone Identification Assistant loaded ===
server(8080).    start web interface, then open http://localhost:8080
start.          console interface
demo.           worked example
run_tests.      run automated tests

?-
```

2. At the `?-` prompt type the following, **including the full stop**, and press Enter:

```
server(8080).
```

The system replies `Web interface running: open http://localhost:8080` in your browser.

3. Open your web browser and go to **`http://localhost:8080`**. The home page appears (Figure 1).

On Windows the first time you start the server, Windows Defender Firewall may ask whether to allow SWI-Prolog to communicate. Choose **Allow** for private networks. The system only listens on your own computer.

Every Prolog command ends with a full stop. If nothing happens after pressing Enter, type . and press Enter again.

4.2 Stop the system

1. In the terminal type `stop_server(8080) .` to stop the web server.
2. Type `halt .` to leave SWI-Prolog. (Closing the terminal window also stops everything.)

5. A tour of the home page

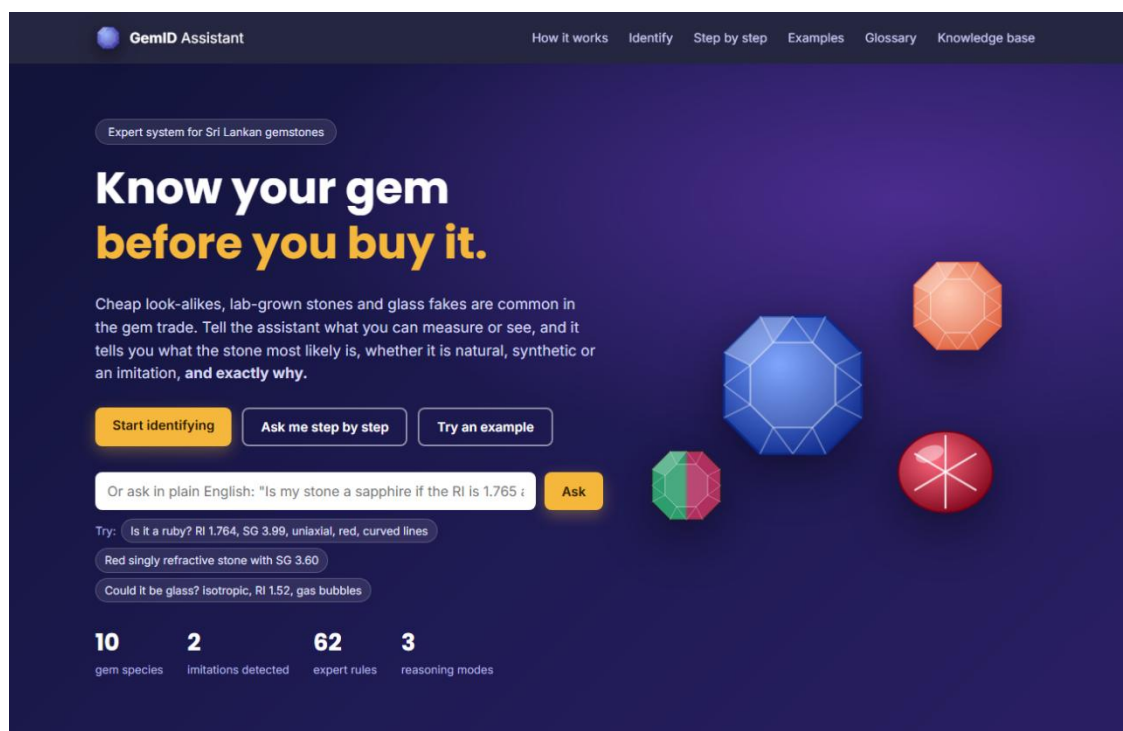


Figure 1. Home page

The top bar links to every part of the system: **How it works**, **Identify** (the input form), **Step by step**, **Examples**, **Glossary** and **Knowledge base**. The introduction offers three buttons: **Start identifying**, **Ask me step by step** and **Try an example**, plus a box for plain-English questions. Scrolling down you will find a summary of what the system tells you, an explanation

of how it works and of the two reasoning methods (Figure 2), the input form, example stones and a glossary of gem terms.

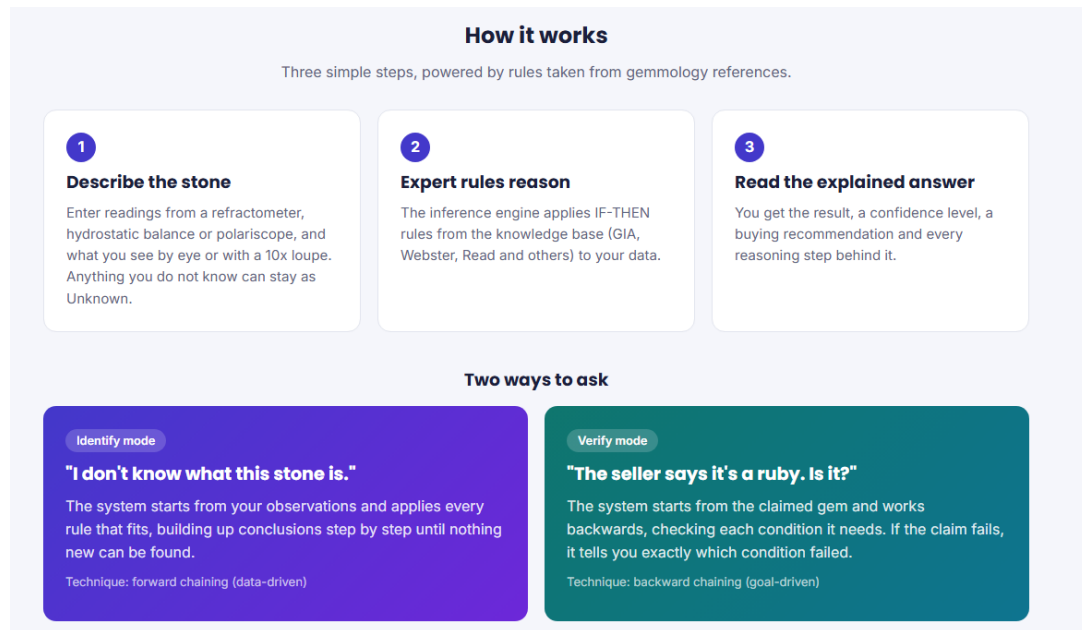


Figure 2. "How it works" section

6. Describing your stone

The form (Figure 3) has two groups of fields. **Every field is optional.** Fill in what you know and leave the rest as *Unknown* / *not checked*. The more you fill in, the more certain the answer. Each field shows what the property means (grey text) and how to measure it (**HOW**).

Describe your stone

Every field is optional. The more you fill in, the more confident the answer.

1 Instrument readings

Refractive index (RI)

How strongly the stone bends light. Each gem has its own range.

☐ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)

How heavy the stone is compared with the same volume of water.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00). [See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

2 What you can see

Colour

The main body colour of the stone. Pick the closest swatch.

☒ Not checked
 ☐ Red
 ☐ Pink
 ☐ Orange
 ☐ Yellow
 ☐ Green
 ☐ Blue
 ☐ Purple
 ☐ Brown
 ☐ Black
 ☐ White
 ☐ Colourless

HOW Look at it in daylight against a white background.

Optical effect

Special light effects. Pick the picture that looks like your stone.

☒ Not checked
 ☐ None
 ☐ Star
 ☐ Cat's-eye
 ☐ Colour change
 ☐ Moonstone glow

HOW Look under a single light source and rotate the stone.

Inclusions

Tiny features inside the stone that reveal how it formed. Pick the sketch closest to what you see.

☒ Not checked
 ☐ Clean
 ☐ Silk
 ☐ Crystals
 ☐ Fingerprints
 ☐ Gas bubbles
 ☐ Curved lines
 ☐ Swirl marks
 ☐ Discoid fractures

HOW Examine it with a 10x jeweller's loupe.

Identify my stone

Find the best match using every rule in the knowledge base.

Check a claim

Test whether the stone could be one particular gem.

Ask me step by step

The system asks only the questions it needs, and you can ask it why.

[Clear the form](#)

Figure 3. The input form: instrument readings with normal ranges, and picture choices for the optic character and for what you can see

Table 2. Input fields

Field	What it is	How to obtain it	Tips
Refractive index (RI)	How strongly the stone bends light. Each gem has its own range.	Gem refractometer with contact liquid; use the highest reading.	Enter three decimals, e.g. 1.765. Allowed range 1.3-3.0. If the scale shows no shadow edge at all, tick Over the limit (no reading) instead. Leave it blank if you have no refractometer.
Specific gravity (SG)	How heavy the stone is compared with the same volume of water.	Hydrostatic balance: $SG = \text{weight in air} / (\text{weight in air} - \text{weight in water})$.	Enter two decimals, e.g. 4.00. Allowed range 1.0-8.0.
Optic character	Whether light passes as one ray (singly refractive) or is split in two (doubly refractive).	Polariscope: rotate the stone between crossed filters. A conoscope (or a glass sphere held over the stone) shows the interference figure.	Click the picture that matches what you see: <i>Stays dark</i> (singly refractive), <i>Blinks light/dark</i> (doubly refractive, figure not checked), <i>Dark cross</i> (uniaxial) or <i>Curved bars</i> (biaxial).
Colour	The main body colour.	Look in daylight against a white background.	Click the swatch closest to your stone.
Optical effect	Star, cat's-eye, colour change or moonstone glow.	Look under a single light source and rotate the stone; compare daylight and lamp light for colour change.	Click the picture that looks like your stone. Choose <i>None</i> if you looked and saw no effect.
Inclusions	Tiny features inside the stone that reveal how it formed.	Examine with a 10x jeweller's loupe.	Click the sketch closest to what you see. <i>Clean</i> means you looked and saw nothing inside.

Optic character, colour, optical effect and inclusions are chosen by clicking a **picture tile**; the chosen tile is outlined in blue. Leave the **Not checked** tile selected if you did not look. Hover over a tile to see its full description.

6.1 Instant checks on your readings

As soon as you type an RI or SG value, a short note appears under the box (Figure 4):

- **Green "Fits: ..."** lists the gems whose range includes your reading.
- **Orange warning:** the value is unusual (no gem in the system has it, or it is above what most refractometers can read). Check your measurement.
- **Red error:** the value is impossible (outside the allowed range). It appears when you leave the box, and the form will not be sent until you correct it.

Under each box a **Normal range** line shows the range of readings covered by the system (for example RI 1.515 - 1.990 for gems). The link **See the range for each gem** opens the ranges of every gem in a new tab, so your form is kept.

Ticking **Over the limit (no reading)** greys out the RI box and shows which gems read above the refractometer's limit (zircon, cubic zirconia and some garnets).

1 Instrument readings

Refractive index (RI)
How strongly the stone bends light. Each gem has its own range.

1.72

✓ Fits: Spinel, Glass.

☐ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)
How heavy the stone is compared with the same volume of water.

9

⚠ SG must be between 1.0 and 8.0.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00).
[See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

1 Instrument readings

Refractive index (RI)
How strongly the stone bends light. Each gem has its own range.

e.g. 1.765

✓ Over the limit: fits Garnet, Zircon, Cubic zirconia.

☒ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)
How heavy the stone is compared with the same volume of water.

1.9

⚠ Unusual reading: no gem in the knowledge base has an SG of 1.9. You can still continue, but please check the reading.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00).
[See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

Figure 4. Readings checked as you type (top) and the "Over the limit" option (bottom)

Below the fields are three action panels: **Identify my stone**, **Check a claim** (choose a gem in the list, then **Verify**) and **Ask me step by step**. **Clear the form** empties all fields.

7. Identifying a stone

7.1 Steps

1. Fill in the fields you know (Section 6).
2. Press **Identify stone**.
3. Read the result page (Figure 5).

7.2 Reading the result

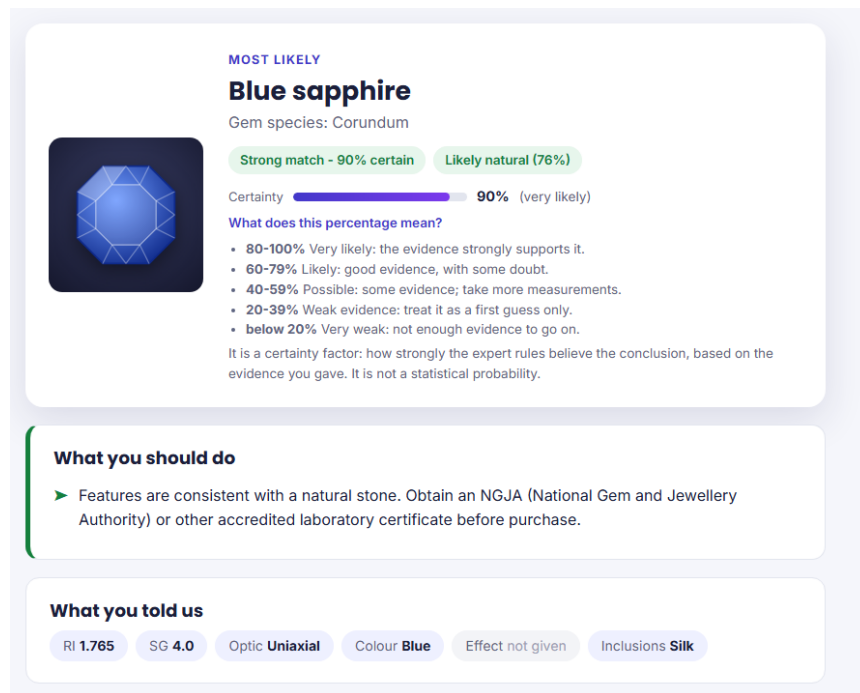


Figure 5. Result for a natural blue sapphire, with the certainty scale opened

Table 3. Parts of the result page

Part	Meaning
Title (e.g. <i>Blue sapphire</i>)	The most likely identification, with the species underneath.
Confidence badge	Strong match (green): optic character, RI and SG all agree. Possible match (amber): data is incomplete or fits more than one gem. No match (grey): nothing in the knowledge base fits.
Certainty bar	How strongly the expert rules believe the conclusion, with a word that explains it: very likely (80-100%), likely (60-79%), possible (40-59%), weak evidence (20-39%), very weak (below 20%). Click What does this percentage mean? to see the scale. It is a certainty factor, not a statistical probability.
Origin badge	Likely natural (green), Synthetic or laboratory test recommended (amber), Imitation or Glass or synthetic (red), Not determined (grey, usually because no inclusions were reported). A Probably heat-treated badge appears when heat indicators were seen.
What you should do	Practical recommendations.
What you told us	Your inputs; fields you left empty show "not given".
Reasoning diagram	A picture of how the rules fired (Section 7.3).
How the system reached this	Each reasoning step in plain English, with the rule number, its certainty, the expert knowledge behind it and its source.

Part	Meaning
About (gem)	Reference facts about the identified gem.
Technical trace	Click to open the full list of facts and fired rules (for gemmologists and examiners).
Buttons	Edit my inputs returns to the form with your values filled in; Print / Save as PDF prints the result (Section 12); Start a new stone clears the form.

7.3 Using the reasoning diagram

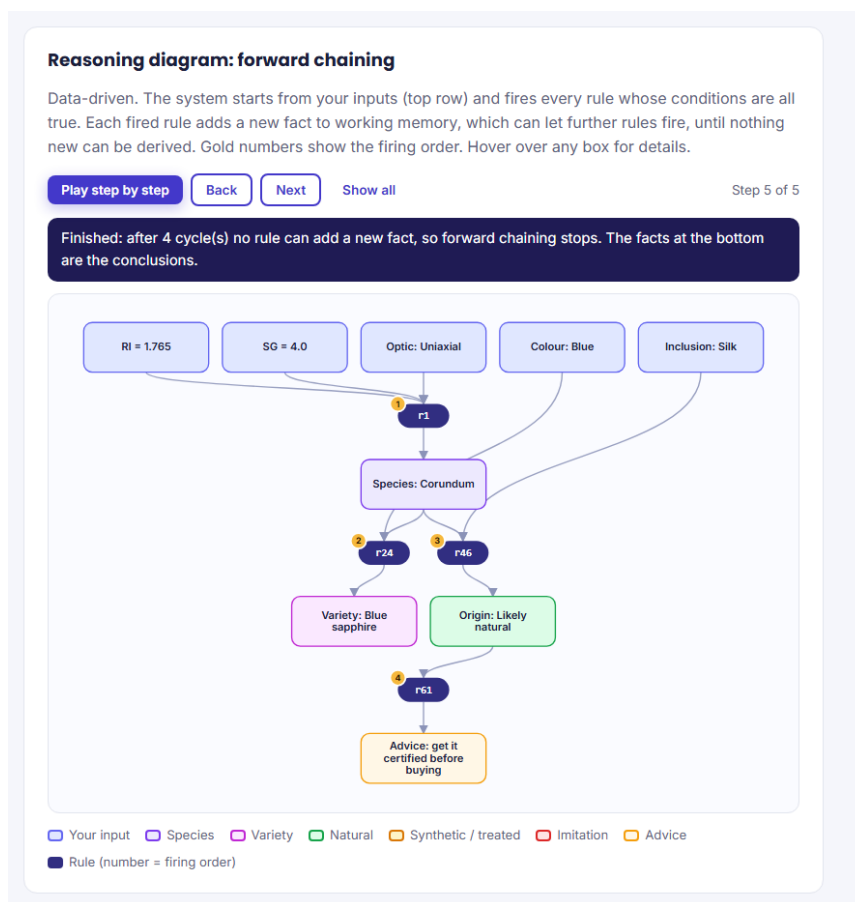


Figure 6. Forward-chaining diagram

Your inputs are the blue boxes at the top. Each dark oval is a rule; its gold number shows the order in which it fired. Arrows show which facts made the rule fire and which new fact it added. Use the buttons:

- **Play step by step:** replays the reasoning automatically, one rule at a time, with a caption explaining each step.
- **Back / Next:** move one step at a time yourself.
- **Show all:** shows the complete diagram again.

Hover the mouse over any box to see its full details.

7.4 When some data is missing

If you leave out the RI, SG or optic character, the system cannot confirm a species. Instead it lists **Ranked possibilities** (Figure 7): every gem consistent with what you entered, ordered by certainty, with the evidence for each.

Above the list, the yellow **Most useful next measurement** box tells you which missing measurement would narrow the list down best, and what each candidate would read. For example, for a red, singly refractive stone it advises measuring the refractive index, because spinel reads 1.712-1.736 and garnet 1.730-1.890.

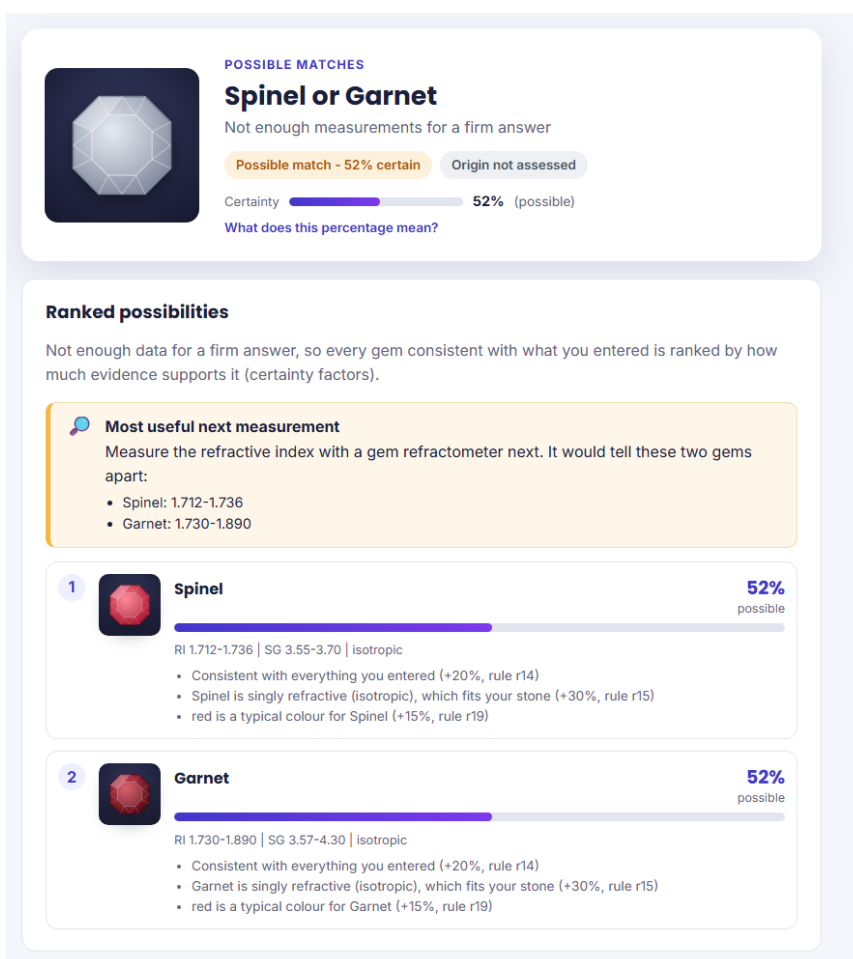


Figure 7. Ranked possibilities and the most useful next measurement for a red, singly refractive stone

8. Checking a claim (Verify)

1. Fill in the fields you know.
2. In the **Check a claim** panel choose the gem the seller claims, e.g. *Ruby*.

3. Press **Verify**.

The answer is a clear **Yes** (green tick) or **No** (red cross). For a **No**, the page shows the **main reason**, what your data points to instead, and a **Why not?** list tracing the failure back through the rules (Figure 8). The backward-chaining diagram shows the question at the top, the rules tried below it, and each condition marked proved (green), failed (red) or never checked (dashed grey).

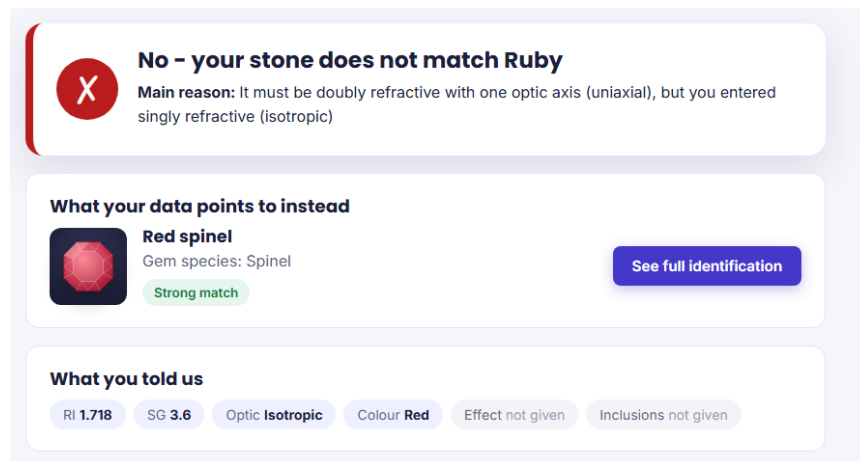


Figure 8. A failed verification: a red spinel offered as a ruby

9. Step-by-step consultation

1. Click **Step by step** in the top bar (or **Ask me step by step** on the home page).
2. Choose what you want to find out. *Find the species* lets the system test one gem hypothesis after another. You can also choose a particular gem.
3. Answer each question and press **Answer**, or press **I don't know**. The bar at the top shows how far you are ("Question 2 of at most 6").
4. Made a mistake? Press **Back** to return to the previous question and change your answer.
5. Open **Why are you asking this?** at any time to see which rule the system is testing and why it needs the answer (Figure 9).
6. When the system has enough information it shows the answer, how many questions it needed, and a diagram of its reasoning. **Change my last answer** takes you back one step.

Question 2 of at most 6

QUESTION 2 testing rule r1

What is the refractive index (RI) of your stone?

Refractive index (RI)
How strongly the stone bends light. Each gem has its own range.

☐ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81.
[See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Answer **I don't know** [← Back](#)

▼ **Why are you asking this?**

1. I am trying to find out: is it Ruby? Rule r21 says: IF Species: Corundum AND Colour: Red AND NOT Effect: Star THEN Variety: Ruby.
2. To use that rule I first need to establish: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
3. To test rule r1 I need to know: the refractive index.

Figure 9. A consultation question with the progress bar, the Back button and the explanation

The system asks only what it needs. For example, when checking for ruby, answering *Isotropic* to the first question ends the consultation immediately, because a ruby must be doubly refractive.

10. Asking in plain English

Type a question into the box on the home page and press **Ask**. The system shows what it understood (Figure 10) and then answers. Examples:

- *Is my stone a sapphire? RI 1.765, SG 4.00, uniaxial and blue*
- *What is a singly refractive red stone with SG 3.60?*
- *Could it be glass? isotropic, RI 1.52, gas bubbles*
- *Blue stone, RI over the limit, SG 4.70, uniaxial*

Understood words include RI / refractive index, "over the limit", SG / specific gravity, singly or doubly refractive, uniaxial, biaxial, colour names, star, cat's eye, colour change, adularescence, silk, crystals, fingerprints, gas bubbles, curved lines, swirl marks, discoid fractures, clean, and gem names. Words the system did not use are listed so you can rephrase.

You asked

"Is my stone a sapphire? RI 1.765, SG 4.00, uniaxial and blue"

The grammar (a Prolog DCG) understood your question as:

Question **Is it Blue sapphire?**
RI **1.765**
SG **4.0**
Optic **Uniaxial**
Colour **Blue**

Yes - your stone is consistent with Blue sapphire

Every condition the expert rules require for Blue sapphire is met by what you entered.

Certainty **90%** (very likely)

What does this percentage mean?

A lab certificate is still recommended before buying.

Figure 10. A plain-English question and how it was understood

11. Worked examples

The **Example stones to try** section of the home page has 15 ready-made stones (Figure 11). Press **Try it** on any card to copy its readings into the form, then press **Identify stone** or **Verify**.

Example stones to try

Press "Try it" to load a stone's readings into the form, then choose Identify or Verify.



Blue sapphire

RI	1.765
SG	4.00
Optic	Uniaxial
Colour	Blue
Inclusion	Silk

EXPECTED Blue sapphire, likely natural

Try it



Synthetic ruby

RI	1.764
SG	3.99
Optic	Uniaxial
Colour	Red
Inclusion	Curved growth lines

EXPECTED Ruby, synthetic (flame fusion)

Try it



Star sapphire

RI	1.765
SG	4.00
Optic	Uniaxial
Colour	Blue
Effect	Star (asterism)
Inclusion	Silk

EXPECTED Star sapphire, likely natural

Try it



Padparadscha

RI	1.766
SG	4.00
Optic	Uniaxial
Colour	Orange
Inclusion	Fingerprints / healed fractures

EXPECTED Padparadscha, likely natural

Try it









Figure 11. Example stones

To test every gem in the system with your own inputs, use the readings in **Appendix A**: each gem's facts, a sample reading to type, and the result the system gives.

11.1 Example 1: a natural blue sapphire

Input	RI 1.765, SG 4.00, Uniaxial, Blue, Silk
Result	Blue sapphire (species corundum) - Strong match, 90% certain (very likely); origin Likely natural (76%) .
Recommendation	Obtain an NGJA or other accredited laboratory certificate before purchase.
Why	Rule r1 identified corundum from the optic character, RI and SG; rule r24 named the blue variety; rule r46 used the rutile silk as evidence of natural origin.

11.2 Example 2: a glass imitation

Input	RI 1.60, Isotropic, Gas bubbles (SG not measured)
Result	Imitation: Glass - 85% certain; origin Imitation (not a natural gemstone) (Figure 12).
Recommendation	This is an imitation with little gem value; do not pay gemstone prices.

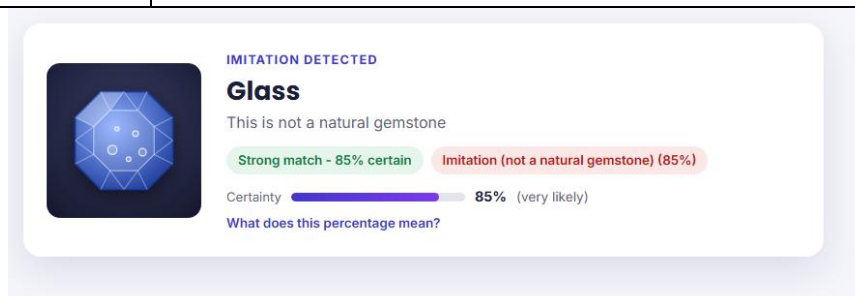


Figure 12. A glass imitation is detected

11.3 Example 3: checking a claim that fails

Input	RI 1.718, SG 3.60, Isotropic, Red; claim: <i>Ruby</i>
Result	No - the stone does not match ruby. Main reason: a ruby must be doubly refractive (uniaxial) but the stone is singly refractive. The data points to Red spinel instead (Figure 8).

11.4 Example 4: a refractometer reading over the limit

Input	RI Over the limit (ticked), SG 4.70, Uniaxial, Blue, Crystals
Result	Blue zircon (species zircon) - Strong match, 65% certain (likely); origin Likely natural (Figure 13).
Why	Only zircon, cubic zirconia and some garnets read above the refractometer's limit; of those, only zircon is uniaxial with an SG of about 4.7. The certainty is lower than for a measured RI because the exact value is unknown.

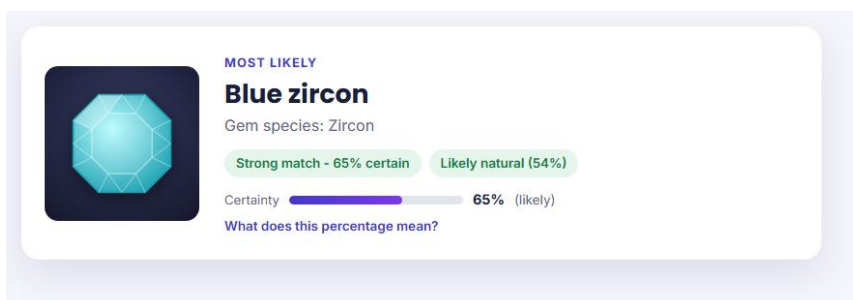


Figure 13. A blue zircon identified with an RI over the refractometer limit

12. Printing or saving a result

Every result page has a **Print / Save as PDF** button at the bottom. It opens your browser's print window: choose a printer, or choose **Save as PDF** as the printer to keep a file. The printout (Figure 14) has a header with the date and time and a reminder that it is a first opinion, leaves out the menu and buttons, and includes the full diagram and technical trace. You can take it to a seller or a gem laboratory.

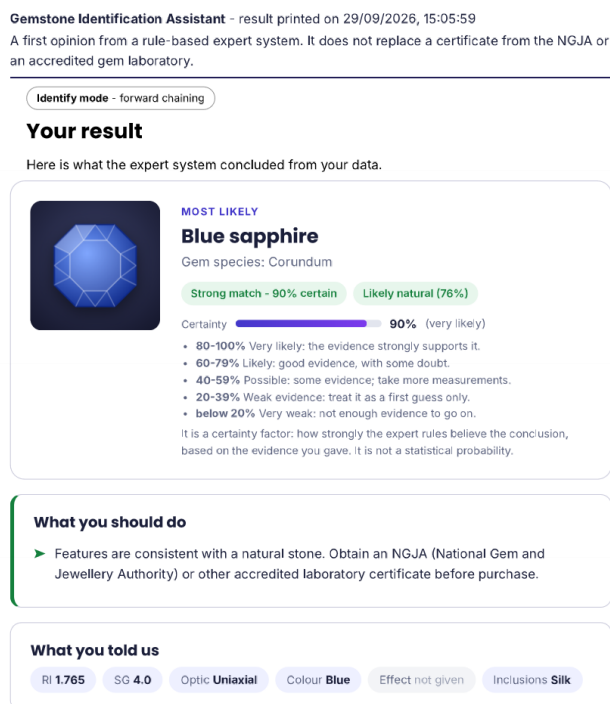


Figure 14. First page of a printed result

13. Your recent stones

The system remembers the last eight results you looked at. They appear at the top of the form on the home page under **Your recent stones** (Figure 15). Click one to open its result again, or press **Clear list** to remove them.

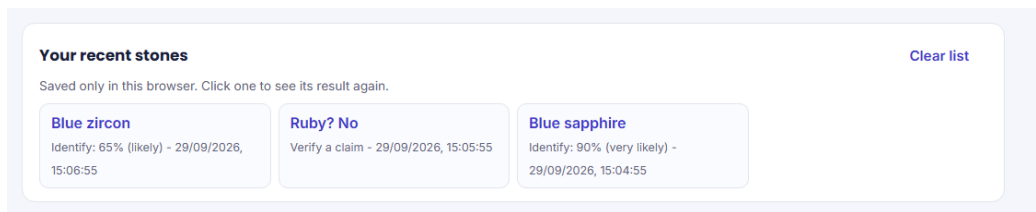


Figure 15. Your recent stones on the home page

The list is stored only in your own browser on your computer; nothing is sent anywhere. It will be empty in a private window, in a different browser, or after clearing your browsing data.

14. Using the system on a phone or tablet

The pages adapt to small screens (Figure 16). On a narrow screen the links in the top bar are replaced by a **menu button** (three lines) in the top right corner: tap it to open the menu, and tap a link or press it again to close. To use a phone on the same network, start the server on the computer and open `http://<computer's IP address>:8080` on the phone; Windows Firewall must allow SWI-Prolog on private networks.

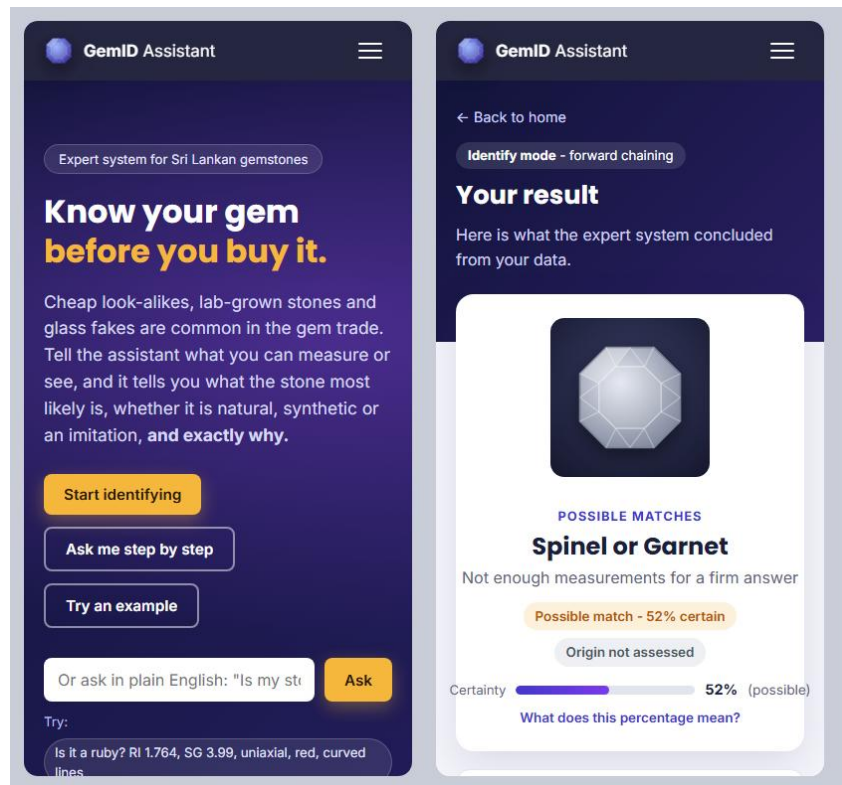


Figure 16. Home page and a result on a phone-sized screen

15. Troubleshooting

Table 4. Common problems and solutions

Problem	Solution
"swipl" is not recognized as an internal or external command"	SWI-Prolog is not on the PATH. Re-run the installer and select "Add swipl to the system PATH", then open a new terminal. Alternatively run it with its full path: "C:\Program Files\swipl\bin\swipl.exe" main.pl.
Error "port 8080 is already in use" (or similar) after <code>server(8080) .</code>	Another program uses that port. Start on a different port, e.g. <code>server(8081) .</code> , and open <code>http://localhost:8081</code> .
Browser shows "This site can't be reached"	The server is not running. Check that you typed <code>server(8080) .</code> with the full stop, and that the terminal window is still open. Check the port number in the address.
Nothing happens after typing a command	Prolog is waiting for the full stop. Type <code>.</code> and press Enter.
Error messages about files not found when running <code>swipl main.pl</code>	The terminal is not in the <code>gem_expert</code> folder. Use <code>cd</code> to move into it, or open the terminal from that folder.

Problem	Solution
"Please check your input" page, or a red message under a box	A value is outside the allowed range (RI 1.3-3.0, SG 1.0-8.0) or is not a number. Correct it (or press Go back and fix it) and try again.
The RI box is greyed out	Over the limit (no reading) is ticked. Untick it to type an RI value.
"Your recent stones" is missing or empty	It only appears once you have looked at a result in this browser. It is empty in private windows, in another browser, or after clearing browsing data.
The print button prints extra pages or no colours	In the print window, turn on Background graphics and choose Save as PDF or your printer; the result normally fits on a few pages.
Result says "We could not identify this stone"	Re-check the readings (RI usually has three decimals, SG two) and the optic character. The stone may also be outside the system's scope; seek a laboratory test.
Changes to the program files do not appear	In the Prolog terminal type <code>make .</code> to reload changed files, then refresh the browser (Ctrl+F5). Or stop and restart the system.
Windows Firewall message on first start	Choose Allow for private networks.
The page fonts look plain	You are offline; the page falls back to standard fonts. Everything still works.

16. Console mode (without a browser)

If the web interface cannot be used, the same functions are available in the terminal.

Step 1. Start SWI-Prolog with `swipl main.pl`.

Step 2. Type `start.` A menu appears:

```

What would you like to do?

  1. Identify a stone - "I don't know what it is"      (forward chaining)
  2. Check a claim   - "The seller says it's a ruby"   (backward chaining)
  3. Step by step    - asks only what it needs; type WHY at any question
  4. Ask in English  - e.g. is it a ruby if the RI is 1.765 and it is red
  5. Run the demo    - natural blue sapphire example
  6. Quit

Choose 1-6:

```

Step 3. Type a number and press Enter (no full stop is needed inside the menu). Answer each question by typing a number or a value; press Enter on its own if you do not know. For the RI, type `OTL` if the refractometer shows no reading (over the limit).

Step 4. In step-by-step mode, type `why` at any question to see why it is being asked:

```
What is the optic character?
  0. Don't know
  1. Isotropic
  2. Uniaxial
  ...
> why
WHY I AM ASKING:
- I am trying to find out: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in
  Corundum range AND SG in Corundum range THEN Species: Corundum.
- To test rule r1 I need to know: the optic character.
```

Step 5. After each result you can type `y` to see the technical trace.

Step 6. Choose `6` to leave the menu, then type `halt`.

The command `demo.` runs the worked blue sapphire example directly.

17. Running the automated tests

Step 1. Start SWI-Prolog with `swipl main.pl`.

Step 2. Type `run_tests.` The last line should read:

```
% All 47 tests passed
```

The rule base can also be exported to a spreadsheet file with `export_rules_csv('rules_table.csv')`.

18. Quick reference

Table 5. Commands

Command (type at the <code>?-</code> prompt)	Effect
<code>swipl main.pl</code>	(in the terminal) Load the system.
<code>server(8080).</code>	Start the web interface at <code>http://localhost:8080</code> .
<code>stop_server(8080).</code>	Stop the web interface.
<code>start.</code>	Start the console interface.
<code>demo.</code>	Run the worked blue sapphire example.

Command (type at the ?- prompt)	Effect
run_tests.	Run the 47 automated tests.
make.	Reload program files after editing them.
halt.	Leave SWI-Prolog.

Table 6. Web pages

Address	Page
http://localhost:8080/	Home page, form, examples, glossary
http://localhost:8080/consult	Step-by-step consultation
http://localhost:8080/rules	Knowledge base: gem facts and all rules

Appendix A. Gem property facts for testing

This table lists the facts the system holds for every material in its knowledge base, so the system can be tested with any gem. For each one:

- **RI** and **SG** are the reading ranges the rules use. The system also accepts readings slightly outside them ($RI \pm 0.005$, $SG \pm 0.03$) to allow for measurement error, but with lower certainty.
- **Try this reading** is a sample input from the middle of the ranges. Enter it in the form (Section 6) and press **Identify stone**. For "RI over the limit", tick **Over the limit (no reading)** instead of typing an RI.
- **System result** is what the system concludes for that reading, with its certainty. These results were produced by running the system on each sample.
- **Source** is the reference the values were taken from (full references are in the project report).

Adding a **colour** from the list names the variety (for example corundum + red gives *Ruby*, chrysoberyl + colour change gives *Alexandrite*), and adding an **inclusion** lets the system judge whether the stone is natural or synthetic. Trying values *outside* the ranges, or a different optic character, shows how the system rejects a gem and explains why.

Table A1. Gem property facts, sample readings and results

	Material and colours	RI	SG	Optic	Mohs	Try this reading	System result	Source
	Corundum Red, Pink, Orange, Yellow, Green, Blue, Purple, Colourless	1.757 - 1.780	3.95 - 4.10	Uniaxial	9	RI 1.769, SG 4.03, Uniaxial	Corundum (95%)	GIA Gem Encyclopedia
	Spinel Red, Pink, Orange, Blue, Purple	1.712 - 1.736	3.55 - 3.70	Isotropic	8	RI 1.724, SG 3.63, Isotropic	Spinel (90%)	Webster, Gems
	Garnet Red, Orange, Green, Purple, Brown	1.730 - 1.890	3.57 - 4.30	Isotropic	7	RI 1.770, SG 3.93, Isotropic	Garnet (90%)	Webster, Gems

	Material and colours	RI	SG	Optic	Mohs	Try this reading	System result	Source
	Chrysoberyl Yellow, Green, Brown	1.740 - 1.760	3.70 - 3.78	Biaxial	8.5	RI 1.750, SG 3.74, Biaxial	Chrysoberyl (95%)	GIA Gem Encyclopedia
	Zircon Red, Orange, Yellow, Green, Blue, Brown, Colourless	1.810 - 1.990	4.60 - 4.80	Uniaxial	7.5	RI over the limit, SG 4.70, Uniaxial	Zircon (77%)	Gemdat.org
	Topaz Pink, Orange, Yellow, Blue, Brown, Colourless	1.606 - 1.644	3.49 - 3.57	Biaxial	8	RI 1.625, SG 3.53, Biaxial	Topaz (95%)	GIA Gem Encyclopedia
	Tourmaline Red, Pink, Yellow, Green, Blue, Brown, Black	1.614 - 1.666	3.00 - 3.26	Uniaxial	7.5	RI 1.640, SG 3.13, Uniaxial	Tourmaline (90%)	GIA Gem Encyclopedia
	Quartz Pink, Yellow, Purple, Brown, Colourless	1.540 - 1.558	2.60 - 2.70	Uniaxial	7	RI 1.549, SG 2.65, Uniaxial	Quartz (90%)	Read, Gemmology
	Feldspar White, Colourless	1.515 - 1.530	2.55 - 2.63	Biaxial	6	RI 1.523, SG 2.59, Biaxial	Feldspar (90%)	Read, Gemmology
	Beryl Pink, Yellow, Green, Blue, Colourless	1.564 - 1.602	2.63 - 2.92	Uniaxial	7.5	RI 1.583, SG 2.78, Uniaxial	Beryl (90%)	Gemdat.org
	Glass Made in any colour	1.440 - 1.770	2.20 - 4.50	Isotropic	5.5	RI 1.605, SG 3.35, Isotropic, gas bubbles	Glass (85%)	International Gem Society
	Cubic zirconia Usually colourless; also dyed colours	2.150 - 2.180	5.60 - 6.00	Isotropic	8.5	RI over the limit, SG 5.80, Isotropic	Cubic zirconia (95%)	GIA Gem Encyclopedia

These values are for testing the system. Real stones vary; always rely on a certificate from the NGJA or an accredited laboratory for a purchase.